Mustafa Mehmood

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Experience

Applied Scientist — London, UK
Thomson Reuters — *June 2025 – Present*

- Hired as 6-month intern initially; contract extended based on performance to take full-time Applied Scientist responsibilities.
- Building tool and retrieval-augmented multi-agent systems in collaboration with legal experts to extract and assess risks across high-stakes legal documents supporting M&A transactions, identifying 90%+ of all potential risks.
- Designed an adversarial evaluation pipeline for agentic solutions with 90%+ failure detection rate (0.78 Cohen's Kappa vs. expert grades). Eliminated slow manual evaluation bottlenecks, cutting the solution development feedback loop from days to hours.
- Fine-tuned LLMs for specialized legal use cases including long-form contract drafting, customization, and summarization.

LLM Data Integrity Specialist — Nottingham, UK
Outlier AI — *May 2024 – May 2025*

- Improved tool usage and function-calling capabilities of advanced reasoning LLMs through RLHF training pipelines, enabling models to infer, chain, and autonomously execute multi-step complex actions from high-level user intent, powering AI solutions for tech firms including Google and Meta.
- Curated and cleaned training datasets alongside domain experts, achieving a 20% improvement in instruction adherence.

Data Scientist Intern — Selangor, Malaysia
Department of Neuroscience, University of Nottingham Malaysia — *June 2021 - Aug 2023*

- Engineered feature extraction pipelines to transform raw EEG and eye-tracking clinical data into structured features: coherence, Hjorth parameters, and spectral power bands for EEG; gaze patterns and fixation analysis for eye-tracking.
- Trained predictive models for neurological and mental health disorder diagnosis, achieving 80% accuracy for Parkinson's and 82% for Schizophrenia from EEG signals, and 92% for Autism from eye-tracking data.

Education

University of Nottingham — Nottingham, UK
MSc *Artificial Intelligence* - ***Grade: Distinction*** — *Sept 2023 – Dec 2024*
BSc *Computer Science* ***(Hons)*** — *Sept 2020 – Aug 2023*
Key Modules: Machine Learning, Computer Vision, Natural Language Processing, Big Data, Information Visualisation

Key Projects

Sherlock-AI: Character Emulation with LLMs — Website Repo

- Designed a resource-efficient, transferable character emulation pipeline: fine-tuned a quantized LLaMA model with *LoRA* for Sherlock Holmes emulation, automating dialogue extraction and synthetic data generation from public domain texts.
- Implemented ReAct-style reasoning for multi-step deductive analysis with dual inference support (OpenRouter API and local GGUF), optimizing for performance and cost across cloud and local setups.
- Built a full-stack web application in Next.js, React, and TypeScript with a Node.js/MongoDB backend, featuring an interactive story mode where users play as characters in branching Sherlock Holmes mysteries through structured decisions and free-form actions, with streaming structured output for real-time scene generation and auto-generated context summaries.
- In A/B testing, 72.4% of users favoured the fine-tuned model over baseline for authenticity, with perfect usability scores.

CourseCompanion: RAG-Powered Course Assistant — Repo

- Developed a scalable educational platform enabling professors to create AI teaching assistants that auto-generate knowledge bases from course materials, featuring chat interfaces, FAQ system, and management dashboards; built with Flask and PostgreSQL using RAG with LangChain and Flair for context-aware responses, deployed on DigitalOcean with Google Drive integration.

DeepGaze: Autism Diagnosis with ML and Webcam Eye-Tracking — Demo Video Repo

- Designed approaches for early autism detection from eye-tracking data visualized as images, combining PCA with sklearn classifiers and a custom Keras CNN with transfer learning to achieve 92% accuracy; built a modular Flask web app for webcam-based data collection, visualization, prediction, and model retraining.

Publication

Mehmood, M., Amin, H. U. (2024). Pre-diagnosis for Autism Spectrum Disorder Using Eye-Tracking and Machine Learning Techniques. *Advances in Brain Inspired Cognitive Systems*, Springer Nature Singapore.

Technical Skills

Programming Languages: Python, C, Java, JavaScript, TypeScript, R, Haskell, SQL, MATLAB, HTML/CSS

Libraries & Frameworks: PyTorch, TensorFlow, Hugging Face (Transformers, trl), scikit-learn, LangChain, LangGraph, Pydantic AI, spaCy, vLLM, React, Next.js, Flask, pandas, NumPy, Weights & Biases, Gradio/Streamlit, Claude Agent SDK

Tools & Platforms: Git (GitHub, GitLab CI), Docker, AWS (SageMaker, EC2, Lambda, Bedrock), Apache Spark/Databricks, PostgreSQL, MongoDB, Runpod, DigitalOcean, Power BI, Claude Code, Cursor